

Considerations for A\$ \$42 Binder Candidates

Biological Function, Formatting Strategy, and Pre-Experimental Concerns

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1. Expected Biological Functions of the 7 Accepted Binders

1.1 What These Binders Target

All 7 accepted binders were designed against the **HHQK epitope** (Y10, E11, H13, H14, Q15, K16) on an A β 42 trimer extracted from PDB 9CO4 (receptor-bound conformation 1). This epitope region is functionally significant:

- **H13/H14** coordinate Cu^{2+} and Zn^{2+} ions — metal binding at these residues drives A β aggregation and reactive oxygen species (ROS) generation.
- **K16** sits at the boundary of the hydrophobic core (KLVFF_{17-21}) that drives β -sheet stacking in amyloid fibrils.
- **The HHQK motif** mediates heparan sulfate proteoglycan (HSPG) binding on neuronal surfaces, a key pathway for A β cytotoxicity.

The binders are therefore not binding an inert surface — they mask a **functionally active region** of A β 42.

1.2 Realistic Biological Activities

What the binders **CAN** likely do

1. **Block metal-induced aggregation.** By shielding H13/H14, binders prevent $\text{Cu}^{2+}/\text{Zn}^{2+}$ coordination that nucleates toxic oligomers. This is a mechanism no approved antibody specifically exploits.
2. **Cap oligomers and prevent elongation.** The binders were designed against a trimer, so they recognize the oligomeric conformation. Binding sterically blocks further monomer addition, acting as aggregation terminators.
3. **Block A β -cell surface interactions.** Masking the HHQK heparan sulfate binding motif reduces neuronal uptake and membrane disruption by oligomers.
4. **Superior tissue penetration.** At 68–87 amino acids (7–10 kDa), these binders are $\sim 15\times$ smaller than an IgG (~ 150 kDa). This is a genuine advantage for CNS delivery, especially when fused to a TfR1-binding module for BBB transcytosis.

What the binders **CANNOT** do

1. **Disaggregate preformed fibrils.** Fibril disassembly requires breaking hundreds of backbone H-bonds along the cross- β spine. No small binder has the thermodynamic leverage for this. Even lecanemab does not truly disaggregate mature fibrils — it preferentially binds soluble protofibrils.
2. **Recruit immune clearance.** Without an Fc domain, there is no antibody-dependent cellular phagocytosis (ADCP), no complement activation, and no microglial recruitment for plaque clearance.
3. **Long serum half-life (unmodified).** Small proteins are cleared renally in minutes to hours versus weeks for IgG. This is addressable via engineering (see Section 2).

1.3 Comparison with Lecanemab

Feature	Lecanemab	These Binders
Target epitope	Conformational protofibril epitope	Linear HHQK epitope on oligomer
Size	~ 150 kDa (IgG)	~ 7 – 10 kDa
Primary mechanism	Fc $\rightarrow$ microglial phagocytosis	Epitope masking (aggregation prevention)
Fibril disaggregation	Minimal (targets protofibrils)	No
Aggregation prevention	Yes (sequesters protofibrils)	Yes (caps oligomers, blocks metal sites)
BBB penetration	Poor ($\sim 0.1\%$ of plasma)	Much better intrinsically
Half-life	~ 5 – 7 days	Minutes–hours (unmodified)
ARIA risk	Yes (Fc-mediated inflammation)	No (major safety advantage)
Production cost	Very high (mammalian cell culture)	Low (bacterial expression)

1.4 Summary of Therapeutic Positioning

These binders are **not lecanemab-like in mechanism**. Lecanemab is fundamentally an immune recruiter that flags A β for destruction by microglia. These binders are **passive blockers** that bind and mask a functional epitope.

The most compelling use case is as the **A β -binding arm of a bispecific** with Tfr1 for BBB transcytosis — a small, brain-penetrant aggregation inhibitor with no ARIA risk. This is a genuinely differentiated therapeutic hypothesis from the antibody approach.

This is comparable effort to Protein A + SEC, which is what industry does with Fc-fused proteins (Protein A alone is never the final step).

2.5 When Would Fc Be Needed?

Fc fusion should only be considered if:

- Experimental data shows that **active immune clearance** of A β deposits is required for efficacy (i.e., passive blocking alone is insufficient).
- You need to move toward clinical development under an **established regulatory framework** (Fc-fusion biologics have a well-trodden FDA path).

However, the field is increasingly shifting toward the view that **soluble oligomers are the toxic species**, not mature plaques — which favors the sequestration/capping approach over immune-mediated plaque clearance.

2.6 Cost Comparison at Research Scale

	Fc fusion (CHO cells)	E. coli + ABD/His-tag
Cell line development	3–6 months, \$50–100K	None
Media cost per liter	\$200–500 (serum-free)	\$5–10 (LB/TB)
Yield	0.5–2 g/L (optimized)	10–100 mg/L (sufficient)
Time to first milligram	4–6 months	2–3 weeks
Endotoxin removal	Not needed	Needed (standard protocol)

At the current stage (binding validation, aggregation assays, cell culture, possibly mouse models), milligrams are needed, not grams. E. coli delivers this in weeks, not months.

3. Pre-Experimental Concerns and Mitigation Strategies

3.1 Concern Priority Matrix

Priority	Concern	Severity	Fix before lab?	Key mitigation
1	Target conformation uncertainty	High	Partially	MD simulation
2	Low MPNN recovery (5/7 designs)	High	Yes	Prioritize s480128
3	Interface Met oxidation (3/7)	Medium	Yes	MPNN M→L/I redesign
4	Hydrophobic-dominated interfaces	Medium	Partially	$\Delta\Delta G$ alanine scan
5	Limited scaffold diversity (4 unique)	Medium	Partially	More BindCraft runs
6	AF2 false-positive rate	Medium	No	Test multiple designs
7	Cross-polymorph reactivity unknown	Low-Med	Yes	ColabFold re-prediction

3.2 Concern 1: Target Conformational Heterogeneity (HIGHEST RISK)

A\$ 42 is an intrinsically disordered peptide that adopts dozens of polymorphs. The binders were designed against one specific trimer conformation from PDB 9CO4. The hotspot residues (9–16) sit at the boundary between the disordered N-terminus (residues 1–8) and the ordered core — this region is **partially flexible even in 9CO4**.

Impact: If the 9CO4 N-terminal conformation is rare under physiological conditions, the binders may fail to recognize real A\$ 42 species.

Mitigation:

- Run **MD simulations** (200–500 ns) on binder–trimer complexes to check interface stability
- Complete the deferred **N-terminus MD project** to characterize epitope dynamics in the free trimer
- Test binding experimentally against both **monomeric and oligomeric** A β 42 — if binding is oligomer-selective, this is a feature (toxic species targeting), not a bug

3.3 Concern 2: Low MPNN Sequence Recovery

Design	MPNN Recovery	Interpretation
s480128_mp13	0.63	Good sequence–structure compatibility
s480128_mp17	0.61	Good sequence–structure compatibility
s120913_mp1	0.41	Moderate — may still fold
s120913_mp2	0.37	Moderate — monitor folding
s716952_mp20	0.33	Weak — higher misfolding risk
s967366_mp11	0.32	Weak — higher misfolding risk
s311665_mp6	0.31	Weak — highest misfolding risk

Recovery < 0.4 means ProteinMPNN finds that the designed backbone does not strongly encode the designed sequence. These designs are more likely to misfold or adopt alternative conformations.

Mitigation:

- Prioritize **s480128 designs** (recovery > 0.6) for first experimental tests
- Validate folding with **CD spectroscopy + thermal melt** before binding assays
- For low-recovery designs: re-run ProteinMPNN at lower temperature (0.05–0.1) to find higher-compatibility sequences

3.4 Concern 3: Methionine Oxidation at Interface

Design	Interface Met count	Risk
s480128_mp13	0	None
s480128_mp17	0	None
s120913_mp1	1	Low
s120913_mp2	2	Moderate
s716952_mp20	3	High
s967366_mp11	3	High
s311665_mp6	3	High

AD brains have elevated oxidative stress. Surface-exposed methionine oxidizes to methionine sulfoxide, which changes side-chain geometry, disrupts binding contacts, creates heterogeneous protein populations, and may increase immunogenicity.

Mitigation:

- For designs with Met ≥ 2 at interface: run **targeted MPNN redesign** fixing all non-Met positions and allowing only M $\rightarrow$ L or M $\rightarrow$ I substitutions (isosteric, oxidation-resistant)
- Test oxidation susceptibility: incubate with 0.1% H₂O₂ for 2 hours, then check binding by SPR
- The **s480128 designs have zero interface Met** — another reason to prioritize them

3.5 Concern 4: Hydrophobic-Dominated Interfaces and Specificity

5 of 7 designs have **zero salt bridges** and only 3–5 H-bonds. Interfaces are dominated by hydrophobic contacts (142–203 pairs). Hydrophobic interfaces tend to be less specific, more prone to non-specific aggregation, and less tolerant of conformational changes.

Only s120913 has salt bridges (2–4 per design), providing electrostatic specificity.

Mitigation:

- Run **Rosetta alanine scanning** ($\Delta\Delta G$) on each design to identify critical interface residues — if binding energy is distributed over many weak hydrophobic contacts rather than concentrated on key residues, specificity is a concern
- Test **off-target binding** against serum proteins (BSA, transferrin, IgG) by SPR — a specific binder should show $<1\%$ cross-reactivity
- Consider computational re-design to introduce 1–2 additional polar contacts at the interface

3.6 Concern 5: Limited Scaffold Diversity

7 designs come from only **5 seeds / 4 unique scaffolds**:

- s120913: 2 designs (mpnn1, mpnn2) — same backbone
- s480128: 2 designs (mpnn13, mpnn17) — same backbone
- s967366: 1 design
- s311665: 1 design
- s716952: 1 design

If a scaffold has a fundamental problem (misfolding, aggregation, off-target binding), both designs from that scaffold fail together. Effective diversity is 5, not 7.

Mitigation:

- Test **at least one design from each scaffold** — do not test both s120913 variants while skipping another scaffold
- Continue running BindCraft to generate more scaffold diversity
- Consider the RFdiffusion $\rightarrow$ ProteinMPNN $\rightarrow$ ColabFold pipeline for orthogonal scaffold generation

3.7 Concern 6: AF2 Validation $\neq$ Experimental Binding

All 7 designs passed BindCraft’s AF2-based filters ($i_pTM > 0.5$, $pLDDT > 0.8$, etc.). However, AF2 is trained on natural proteins and can give false-positive confidence for designed sequences. Literature success rates for computationally designed binders range 15–50% depending on method and target.

Realistic expectation: 2–4 of the 7 designs will actually bind in the lab.

Mitigation:

- This is inherent uncertainty — not fixable computationally
- **Test all designs** (or at least 4–5) to maximize hit rate
- Gene synthesis for all 7 costs ~\$700 total — cheap insurance
- Use **yeast surface display** if available for parallel screening

3.8 Concern 7: Unknown Cross-Polymorph Reactivity

Binders were designed against 9CO4 (conformation 1), but the brain contains multiple A β 42 polymorphs simultaneously. Whether these binders cross-react with other polymorphs (Type II fibrils, Osaka mutant, Iowa mutant) is unknown.

Mitigation:

- Run **ColabFold re-prediction** of binders complexed with A β 42 from other fibril structures (7Q4B, 6SHS, etc.) — check if i_pTM remains high
- This is purely computational and costs only GPU time

4. Recommended Pre-Experimental Action Plan

Stage A: Computational Validation (1–2 weeks on HPC)

1. **MD simulations** on top 3 binder–target complexes (s120913_mp1, s480128_mp13, s311665_mp6)
 - 200–500 ns each in explicit solvent using GROMACS
 - Monitor: interface RMSD, contact persistence, binding free energy (MM-GBSA)
 - **Kill criterion:** if binder dissociates or RMSD >5 Å within 100 ns → discard design
2. **Rosetta $\Delta\Delta G$ alanine scan** on all 7 designs
 - Identifies critical residues and interface robustness
 - 1–2 hours per design on CPU
3. **Cross-polymorph ColabFold prediction**
 - Predict binding to 2–3 other A β 42 fibril structures
 - 30 min per design on GPU
4. **MPNN Met-substitution** for designs with ≥ 2 interface Met
 - Re-design with M→L/I constraint, re-validate with ColabFold

Stage B: Gene Synthesis + Pilot Expression (2–3 weeks)

5. **Order synthetic genes** for 4–5 designs (one per scaffold)
 - Codon-optimized for E. coli
 - With N-terminal His₆-SUMO tag for expression and purification
 - Estimated cost: \$500–700 total
6. **Express and assess folding**
 - SDS-PAGE for expression level
 - SEC for monodispersity (no aggregation)
 - CD spectroscopy for secondary structure (>60% helical expected)
 - Thermal melt (T_m >50°C is reassuring for a designed protein)

Stage C: Binding Validation (1–2 weeks after protein in hand)

7. **SPR or BLI binding assay** against:
 - Monomeric A β 42 (HFIP-treated, freshly dissolved)
 - Oligomeric A β 42 (4°C incubation, 24 h)
 - Off-target controls (BSA, transferrin, IgG)
8. **ThT aggregation inhibition assay**
 - Measure Thioflavin T fluorescence with and without binder
 - Determines if the binder slows A β 42 fibril formation

Stage D: Functional Characterization (if binding confirmed)

9. **Oligomer capping assay** — DLS/SEC to check if binder arrests oligomer growth
10. **Metal competition assay** — test if binder blocks $\text{Cu}^{2+}/\text{Zn}^{2+}$ binding to A β 42 H13/H14
11. **Cell-based neurotoxicity protection** — SH-SY5Y or primary neurons, measure viability with/without binder in the presence of A β 42 oligomers
12. **Bispecific construct assembly** — fuse best A β 42 binder with TfR1 binder + ABD, express as single chain, validate dual binding

5. Recommended Minimum Test Panel

Based on balancing scaffold diversity, MPNN recovery, interface quality, and functional features:

Priority	Design	Scaffold	Rationale
1	s480128_mp13	s480128	Highest MPNN recovery (0.63), zero interface Met, clean metrics. Safest first bet.
2	s120913_mp1	s120913	Highest i_pTM (0.81), 14/18 hotspots, 4 salt bridges. Best binding potential.
3	s716952_mp20	s716952	Unique chain G coverage (6/6 including H13/H14). Distinct epitope engagement.
4	s311665_mp6	s311665	Smallest (68 aa, 7.5 kDa), best for bispecific fusion. Format candidate.
5	s967366_mp11	s967366	Highest pLDDT (0.94), best shape complementarity. Only if budget allows 5th design.

Key principle: Test at least one design from each unique scaffold. Do not test both s120913 variants (mpnn1 + mpnn2) while omitting a different scaffold entirely. Scaffold-level diversity matters more than sequence-level diversity for maximizing the probability of at least one experimental hit.

Report generated for the $A\beta_{42} \times TfR1$ bispecific binder design project.

BindCraft campaign on Frontenac HPC (Queen's University).

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